Lohan Atapattu

Biomedical Engineering Graduate — University of Moratuwa

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Profile

Biomedical Engineering graduate from the University of Moratuwa with hands-on experience in analog and mixed-signal IC design, ASIC development, FPGA-based digital systems, and digital signal processing. Research experience includes biomedical IC design, ear-EEG systems, and biosignal acquisition and processing, alongside industry experience in high-frequency modem development. Contributed to undergraduate IC tapeout initiatives and research publications in biomedical circuits and systems.

Research Interests: Low-power analog and mixed-signal integrated circuits, neural interfaces, biomedical electronics, and implantable medical devices.

Work and Research Experience

1.) FPGA and DSP Engineer – Thakshana Technologies (Pvt) Ltd, Sri Lanka *July 2026 – October 2026*

- Design and verify FPGA-based high-frequency communication modems for military applications, including digital modulation, demodulation, and signal processing modules using VHDL.

2.) Visiting Research Intern – University of Melbourne, Australia linkedin *December 2024 – May 2025*

***John NeuroBionics Lab**, Department of Biomedical Engineering, Faculty of Engineering*

- Conducted research on the design and development of a robotic gripper for assistive biomedical applications.
- Developed an electronic assistive pickup device for individuals with muscle-wasting disorders, focusing on functional design, usability, and system integration.

3.) Teaching Assistant - University of Moratuwa, Sri Lanka *December 2025 - June 2026*

- Supported undergraduate laboratory sessions for EN 1094: Laboratory Practice and EN 2091: Laboratory Practice and Projects, assisting students with experiments and practical engineering activities.
- Supported the EN 4431: Analog IC Design module by assisting with the installation and configuration of Cadence design tools for university use.

4.) Instructor - SkillSurf *March 2025 – Present*

- Contribute to technical education through specialized bootcamps in IoT, processor design, IC design, and related emerging technologies. SkillSurf

5.) Researcher - Biomedical Research in IC Design (BRICD)

- Contribute to research and development in biomedical integrated circuits and electronic systems, focusing on IC-based solutions for biomedical applications. Biomedical Research in IC Design (BRICD)

Selected Lectures and Workshops

Course / Workshop	Description	Year
CMOS Analog IC Design and Simulation	Lectures on CMOS analog IC fundamentals, design methodologies, and circuit-level simulation.	2025
SystemVerilog for ASIC/FPGA Design and Simulation	Lectures on SystemVerilog-based digital design, verification concepts, and simulation workflows.	2025
The Art of Analog CMOS IC Design Workshop	Hands-on workshop covering practical analog CMOS design techniques, design trade-offs, and implementation considerations. Workshop Recording	2025

Education

Institution	Qualification	Period
University of Moratuwa	B.Sc. Engineering (Honours), Biomedical Engineering Transcript **GPA: 3.80 / 4.0** ***(first class)*** ***Dean's List: Semester 1, 2, 4, 6, 7, and 8***	Mar 2021 - June 2026
Ananda College	GCE Advanced Level - Physical Science Stream Z-Score: 2.3086	Jan 2012 - Oct 2020
Association of Chartered Certified Accountants (ACCA)	Financial Reporting, Financial Management, Law, Taxation	Oct 2020 - Present

Relevant Coursework*: Digital and Analog IC Design, Analog Circuit Design, Digital Systems Design, Power Electronics, Modelling and Analysis of Physiological Systems, Signal Processing, Linear Algebra, Applied Statistics, Control Systems, Robotics, Data Structures and Algorithms*

Publications

- "From Circuit Design to Silicon: Exploring the OSIC Design Toolchain and Affordable Chip Tapeouts using Open-source PDKs," E-Carrier (March 2025)
  linkedin.com/...
- **L. Atapattu**, S. Madugalle, I. Nethmal, E. Jayathilaka, A. Herath, K. Wickremasinghe, S. L. Kappel, N. Udayanga, and C. U. S. Edussooriya, "A Dry-Contact Ear-EEG System With Continuous Electrode-Skin Impedance Mismatch Monitoring for Motion Artifact Cancellation Using DRL Stimulus," in **2026 IEEE Biomedical Circuits and Systems Conference (BioCAS)**, **2026**. alphaXiv
- I. Nethmal, **L. Atapattu**, E. Jayathilaka, S. Madugalle, A. Herath, K. Wickremasinghe, S. L. Kappel, N. Udayanga, and C. U. S. Edussooriya, "An Open-Source CMOS ASIC for I/Q Based Electrode-Skin Impedance Mismatch Extraction," in **2026 IEEE International Conference In Microelectronics (ICM)**, **2026**. google drive

Projects

Mixed-Signal IC Design

- **100 KSPS 8-bit Fully Differential SAR ADC – UNIC-CASS** *2024*
  Supervisors: Dr. Nilan Udayanga, Mr. Kithmin Wickremasinghe
  Designed and submitted for tapeout a low-power 8-bit successive approximation register (SAR) ADC operating at 100 KSPS for IoT and wearable/implantable biomedical applications. Focused on circuit design, simulation, and power-area-resolution trade-offs.
  GitHub

Analog IC Design

- **Open-Source CMOS Neuroamplifier Design** *Ongoing*
  Supervisor: Dr. Nilan Udayanga
  Designing a low-power, low-noise CMOS neural recording amplifier inspired by the work of Reid R. Harrison and Cameron Charles, focusing on pseudoresistor-based AC coupling, electrode DC-offset rejection, and amplification of low-amplitude neural signals using an open-source IC design flow.

- **Fully Differential Operational Amplifier Design** *2025*
  Supervisors: Dr. Nilan Udayanga, Dr. Chamira Edussooriya
  Designed and submitted a fully differential operational amplifier for tapeout, targeting low-frequency filter design for EEG acquisition systems. Focused on gain, bandwidth, stability, and power efficiency.
  Project Details

- **Electrode-Skin Impedance Measurement IC for Biomedical Applications** *2026*
  Supervisors: Dr. Nilan Udayanga, Dr. Chamira U. S. Edussooriya
  Designed a low-power analog IC for continuous electrode-skin impedance mismatch monitoring using a driven-right-leg stimulus. Implemented a fully differential active-RC bandpass filter and analog multiplier

circuits for I/Q demodulation using open-source IC design tools and the SKY130 CMOS process.
GitHub , Poster

Digital IC Design

- **Digital IC Design of a Simple Counter Using Cadence Tools** *2025*
  Supervisor: Dr. Subramaniam Thayaparan
  Designed an RTL counter and completed the ASIC implementation flow using Cadence EDA tools, including logic synthesis, technology mapping, floorplanning, placement, and routing.
  Project Files

RF IC Design

- **Ring-VCO-Based Phase-Locked Loop (PLL) Design** *2026*
  Supervisor: Dr. Subramaniam Thayaparan
  Designed a phase-locked loop targeting an operating frequency range of 50–150 MHz using Cadence design tools, with a ring-based voltage-controlled oscillator (VCO) as the core frequency-generation block.
  Project Files

Digital System Design

- **Defense Communications: US MIL-STD-188-110B Baseband Link-Level Simulation** *2026*
  Designed and implemented the FPGA-based baseband modulator in VHDL, including symbol mapping and modulation signal generation, based on the US MIL-STD-188-110B standard.
  MATLAB Reference

- **UART Transceiver Implementation in FPGA** *2024*
  Designed a UART transceiver using Verilog HDL and verified its functionality through simulation and on-hardware testing on a DE0-Nano FPGA board, focusing on reliable serial communication and timing correctness.

- **Floating-Point Unit (FPU)** *2024*
  Designed a 32-bit floating-point unit capable of performing addition, subtraction, multiplication, and division. Implemented arithmetic algorithms at the RTL level and integrated the design on an FPGA platform for functional validation.

Electronic Product Design

- **Wearable Ear-EEG System for Motion Artifact Cancellation** *2026*
  Supervisors: Dr. Nilan Udayanga, Dr. Chamira Edussooriya, Assistant Professor Simon Lind Kappel
  Designed and implemented a wearable dry-contact Ear-EEG acquisition system with continuous electrode–skin impedance mismatch monitoring for motion artifact cancellation. Integrated analog front-end circuitry, impedance sensing, and adaptive digital signal processing to improve EEG signal quality during movement.
  Poster

- **Assistive Robotic Gripper and Pickup Stick Design** *2025*
  Supervisor: Assistant Professor Sam John
  Designed an assistive robotic gripper and pickup-stick device for individuals with muscle-wasting disorders. The project involved 3D mechanical design, electronic product development, actuator and sensor selection, and robotic control concepts, focusing on usability and assistive functionality.
  Project Details

- **EMG Signal Acquisition and Gesture-Controlled Game Interface** *2025*
  Designed and implemented an electromyography (EMG) signal acquisition system involving analog front-end design, signal conditioning, filtering, and feature extraction. Used acquired EMG signals for real-time gesture-based control of a custom Flappy Bird game.
  Project Details

- **Analog Computer Design** *2023*
  Designed and analyzed analog circuits for addition, subtraction, and four-quadrant multiplication using operational amplifiers. Developed schematics and PCB layouts in Altium Designer and mechanical enclosures in SolidWorks.
  GitHub

Biosignal Processing and Machine Learning

- **PCG Signal Classification Using Machine Learning** *2025*
  Developed a phonocardiogram (PCG) signal classification system using datasets from PhysioNet and the Pascal Heart Sound Dataset. Focused on biomedical signal preprocessing, feature extraction, and convolutional neural network (CNN)-based classification of normal and pathological heart sounds.
  GitHub

Volunteering Experience

Sasnaka Sansada 2021 – Present
Pillar Head – Partnership Development
Lead teams and strategic partnership initiatives, overseeing stakeholder engagement and organizational development within a leading Sri Lankan youth volunteering organization.

IEEE Engineering in Medicine and Biology Society (IEEE EMBS) 2024 – 2026
Co-Chair – Brainstorm 2024; Secretary (2024–2025); Councillor (Present)
Led organizational activities for a national biomedical innovation competition involving 50+ teams. Managed chapter administration, grants, reporting, and executive committee coordination.

IEEE Circuits and Systems Society (IEEE CAS) 2025 – Present
Founding Chairperson
Established the first IEEE CAS Student Branch Chapter in Sri Lanka at the University of Moratuwa, promoting academic and professional development in circuits and systems engineering.

Department of Electronic and Telecommunication Engineering 2024 – 2025
Batch Representative (Semesters 4–6)
Represented undergraduate students in academic and administrative matters, coordinating internship processes and departmental communication.

Electronic Club – University of Moratuwa 2025 – 2026
Human Resource Manager
Developed student selection criteria for internships and professional development programs and managed membership records.

Police Cadet Wing – Ananda College 2019
Cadet Sergeant
Led a cadet platoon to achieve Second Place Island-wide at the 2019 National Police Cadet Competition.

Awards and Certifications

First Runner-Up – Spark 2024
Recognized in the spark competition which promotes innovative, sustainability-driven ideas to address global challenges effectively.

Dean's List Honours — Faculty of Engineering, University of Moratuwa (Semesters 1, 2, 4, 6, 7 and 8)

Junior Inventor of the Year — Ananda College, Colombo (2013)

School Colours for Cadet — Ananda College, Colombo (2016)

All-Island 2nd Place — National Police Cadet Corps (2019)

Technical Skills and Tools

Category	**Tools / Technologies**
Analog Circuit Design	LTspice, Ngspice
Digital Systems	Vivado, Intel Quartus Prime, Verilator
PCB Design	Altium Designer, EasyEDA
VLSI Design	xschem, Magic VLSI, Cadence (ASIC flow)
Programming	Verilog, VHDL, SPICE, Python, C++, MATLAB
3D Design	SolidWorks
Machine Learning	PyTorch

References

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